

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	10 July 2026
Team ID	SWTID-2026-6119
Project Name	Credit Card Approval Prediction System
Maximum Marks	4 Marks

Technical Architecture:

The diagram below shows how a user's request flows through the Credit Card Approval Prediction System - from the browser, through the Flask application and preprocessing pipeline, into the trained ML model, and back as a prediction result.

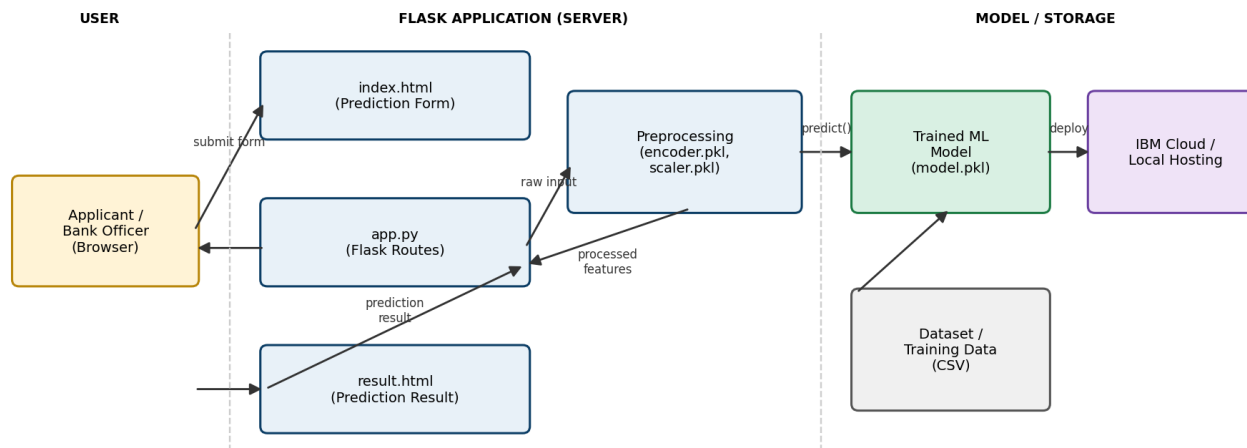


Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	The way users engage with the application	HTML, CSS, Bootstrap, Jinja2 templates (Flask)

S.No	Component	Description	Technology
2	Application Logic	Manages form submissions, routing, and links the UI with the ML model	Python, Flask
3	Data Preprocessing	Transforms categorical values and normalizes numeric fields prior to prediction	Scikit-learn (LabelEncoder / OneHotEncoder, StandardScaler)
4	Machine Learning Model	Generates the Approval or Rejection outcome using processed applicant data	Scikit-learn, XGBoost (best-performing model selected after comparison)
5	Model Persistence	Saves the trained model and preprocessing artifacts so they can be reused during inference	Pickle / Joblib (model.pkl, encoder.pkl, scaler.pkl)
6	Dataset	Past credit card application records used to train and evaluate the model	CSV file (Pandas)
7	File Storage	Holds the dataset, saved model files, and static assets	Local filesystem (project repository)
8	Infrastructure	The environment where the Flask app is hosted and deployed	Local server for development; IBM Cloud for deployment (if required)

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Every major framework in use is open source	Flask, Scikit-learn, XGBoost, Pandas, NumPy, Bootstrap
2	Security	The form performs basic input checks, and no sensitive applicant data is stored past the prediction request	Flask input validation, WTForms (optional)
3	Scalable Architecture	The web layer is decoupled from preprocessing and inference, allowing the model to be retrained or replaced without UI changes	Modular Flask app structure (routes / model / templates)
4	Availability	A lightweight Flask app with few dependencies ensures fast startup and response for demos and evaluation	Flask development server / Gunicorn (for production)

S.No	Characteristics	Description	Technology
5	Performance	Predicting a single applicant record is nearly instantaneous, as the model is pre-trained and loaded only once when the app starts	In-memory model loading via Pickle/Joblib